

A Review of Indirect Time-of-Flight Technologies

Cyrus Bamji^{ID}, Member, IEEE, John Godbaz, Minseok Oh^{ID}, Member, IEEE, Swati Mehta^{ID}, Andrew Payne^{ID}, Sergio Ortiz, Satyadev Nagaraja, Travis Perry, and Barry Thompson, Member, IEEE
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Abstract—Indirect time-of-flight (iToF) cameras operate by illuminating a scene with modulated light and inferring depth at each pixel by combining the back-reflected light with different gating signals. This article focuses on amplitude-modulated continuous-wave (AMCW) time-of-flight (ToF), which, because of its robustness and stability properties, is the most common form of iToF. The figures of merit that drive iToF performance are explained and plotted, and system parameters that drive a camera's final performance are summarized. Different iToF pixel and chip architectures are compared and the basic phasor methods for extracting depth from the pixel output values are explained. The evolution of pixel size is discussed, showing performance improvement over time. Depth pipelines, which play a key role in filtering and enhancing data, have also greatly improved over time with sophisticated denoising methods now available. Key remaining challenges, such as ambient light resilience and multipath invariance, are explained, and state-of-the-art mitigation techniques are referenced. Finally, applications, use cases, and benefits of iToF are listed.

Index Terms—3-D imaging, amplitude-modulated continuous wave (AMCW), CMOS image sensors (CISs), indirect time-of-flight (iToF), modulation contrast (MC), multipath, phase unwrapping, photon-mixing device, quantum efficiency (QE), range imaging, time-of-flight (ToF).

I. INTRODUCTION

SCENE understanding from a 2-D image can be challenging. An object's apparent size depends on its distance from the camera, which is often hard to determine. Many objects of very different shapes can appear the same (e.g., a painting of a ball versus a physical ball), and finally, the same object can appear very different depending on lighting and shadows. Depth cameras facilitate scene [1] and object [2] understanding by providing the distance to the object at each pixel.

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Cyrus Bamji, John Godbaz, Minseok Oh, Swati Mehta, Satyadev Nagaraja, Travis Perry, and Barry Thompson are with Microsoft, Mountain View, CA 94043 USA (e-mail: cybamji@microsoft.com; jgod@microsoft.com; minoh@microsoft.com; swati.mehta@microsoft.com; sanagara@microsoft.com; trperry@microsoft.com; bathomps@microsoft.com).

Andrew Payne was with Microsoft, Mountain View, CA 94043 USA. He is an Independent Consultant on ToF Camera Design, New Zealand (e-mail: andrew.d.payne@live.com).

Sergio Ortiz was with Microsoft, Mountain View, CA 94043 USA (e-mail: seortize@microsoft.com).

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TABLE I

TWO BROAD CATEGORIES OF DEPTH CAMERAS

	Examples	Characteristics
Geometric	Stereo Structured light	Performance relies on accuracy of geometry. Requires baseline
Time of Flight	iToF dToF	Performance dictated by SNR. Requires a transmitter

Indirect time-of-flight (iToF) cameras are a type of depth camera used to produce a depth map where every pixel independently measures the distance to objects in the scene [3]. They can excel in situations when high pixel count, low systematic bias (high accuracy), and low random error (high precision) are simultaneously required [4]–[6]. iToF sensor CMOS process requirements are highly similar to conventional 2-D CMOS image sensors (CISs) and follow a familiar integrate-followed-by-readout scheme.

iToF pixel sizes are shrinking rapidly, approaching competitiveness with global shutter (GS) CIS Pixels [7]. This article focuses on an iToF methodology called amplitude-modulated continuous wave (AMCW) [3], which gives up the freedom to have arbitrary operating waveforms in order to achieve enhanced stability and predictability. AMCW pixel size and performance trends over time are charted in Section IV. Embedded processing algorithms are essential to produce higher quality data. State-of-the-art algorithms for depth are addressed in Section V. As with any other technology, iToF faces challenges, such as sensitivity to ambient light and multipath, and examples of recent advances and mitigation strategies are provided in Section VI. Finally, applications and use cases that showcase the technology are presented in Section VII.

A. Geometric Versus ToF Methods

Depth cameras fall into two broad categories: geometric and time-of-flight (ToF) as shown in Table I. Geometric methods require high mechanical accuracy and generally require a baseline, whereas ToF methods do not depend on geometry nor require a baseline and are hence more mechanically robust and compact [8]. All ToF methods are active methods whose performance is governed by a signal-to-noise ratio (SNR) and affected by ambient light.

B. Types of ToF

There are two main categories of full-field ToF: direct ToF (dToF) and iToF.

In dToF, one or more optical pulses are emitted, and at each pixel, the arrival time of the back-reflected signal is measured. This can be achieved either by analog accumulation, such as in a range-gated system, or by digital detection. In order to avoid being influenced by return signal intensity, the arrival time measurement must trigger with the minimum signal possible, ideally requiring only a single pulse [9]. The development of single-photon avalanche diodes (SPADs) operating in the Geiger mode, which can be triggered by a single photon, has made low-signal digital detection practical. Advanced CMOS nodes with wafer stacking facilitate compact subnanosecond measurement of these arrival events at each pixel [10].

In iToF, time-gated integrations of a light pulse are performed at each pixel in such a way that the accumulated signal value depends on object distance in a manner different from the usual $1/\text{distance}^2$. Using multiple measurements with different distance-dependent functions, it is possible to determine the distance. Due to space limitations, we exclude frequency-modulated continuous-wave (FMCW) systems from discussion in this article.

II. INDIRECT TOF

A. Time-Gated Light Pulse Integration

Unlike radio receivers that have access to the instantaneous radio signal, most image sensors accumulate a photonic signal during an integration time and then periodically read it out in a reset/integrate/read cycle. The accumulated signal is indicative of some value of interest, e.g., photon arrival rate, which corresponds to a quantity of interest such as the intensity of back-reflected light from an object. This intensity can be calculated by dividing the accumulated value by the integration time.

This strategy decouples the number of pixels/detectors from the number of independent readout circuits, typically consisting of amplifiers and analog-to-digital converters (ADCs). Thus, an image sensor with a 10M pixel array can operate with significantly fewer ADCs (e.g., perhaps 3k). Furthermore, since the readout can be performed slowly, the signal paths may be shared between pixels with shared column bit-lines and ADCs at the periphery of the pixel array.

Despite the slowness of the reset/integrate/read cycle, it is still capable of capturing ultrafast events on timescales thousands or even millions of times shorter than the integration time. For example, the capture of fast optical phenomena with slow photographic film dates back to 1931 with the invention of the electronic flash [11].

iToF allows measurement of distance with the standard reset/integrate/readout cycle but requires two fast capabilities:

- 1) a fast light source capable of creating light pulses generally substantially shorter than the distance to be measured;
- 2) a fast signal-gating mechanism (called a shutter) capable of starting/stopping the integration of photocharges in pixels.

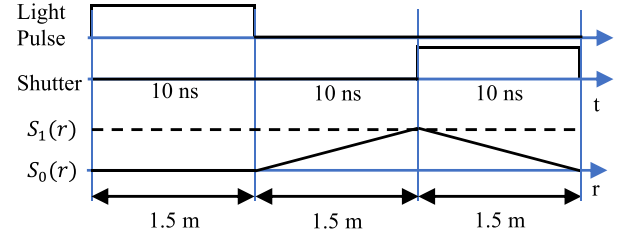


Fig. 1. Time-gated pulse integration.

By synchronizing the light source pulse with the shutter, a time-gated integrated signal $S(r)$ can be generated—this corresponds to the cross correlation of the back-reflected illumination with the shutter response and is dependent on target distance. By using different timings for the light source pulse and shutter, multiple distinct functions $S_i(r)$ can be created, allowing radial distance, r , to be solved for as the solution to an inverse problem.

For example (in the absence of other light and ignoring the reduced signal back-reflected by the object due to increments in the distance), the light pulse and shutter configuration results in a distance-dependent integrated signal value $S_0(r)$ shown in Fig. 1. $S_0(r)$ is the signal produced by integrating light back-reflected by a target at radial distance r . If $S_1(r)$ is the unshuttered signal corresponding to $S_0(r)$ it can be shown that for r between 1.5 and 3 m, the formula

$$r = \frac{10^{-8}c}{2} \left(1 + \frac{S_0(r)}{S_1(r)} \right) \quad (1)$$

holds even when considering the change in object brightness with distance, where c is the speed of light.

While both iToF and range-gated dToF systems can be considered in this manner, iToF can be distinguished by having typical gate duty cycles of 50% versus much lower duty cycles for range-gating—giving range-gating lower collection efficiency and requiring more readouts.

B. Signal Accumulation via Repetitive Shuttering

Because of the short duration, the number of photoelectrons integrated during a single laser pulse is typically extremely small. Measurements for depth require an accumulated signal value of anywhere from around $3e^-$ to $1000e^-$ depending on the application, algorithms, and sensor. Extrapolating the optical power numbers in [12] to a single ~ 10 -ns laser pulse gives a required laser peak power potentially above 10 kW. Since such lasers are expensive, it is beneficial to break the large pulse into many smaller ones (with associated shutters having the same relationship to each pulse). The accumulated signal across this plurality of pulses is the same as that of the single large pulse. Inexpensive lasers are generally < 10 W, so over $> 1k$ pulses may need to be accumulated with $< 1e^-$ of signal captured at each pulse. Thus, shuttering methods used in iToF must corrupt the signal $\ll 1e^-$ at each shuttering cycle.

C. Shutter Types

1) *External Shutters*: As discussed in Section II-B, iToF methods require time-gating the received signal using

synchronized shutters. A liquid crystal shutter with switching speed in the kHz is far too slow to be used for iToF—however, there are faster techniques more suited to iToF.

Since the 1970s, active time-gated image intensifiers have been used to extend imaging range under hazy conditions. Csorba [13] consists of a roughly nanosecond pulsed laser beam illuminating a target and a gated image intensifier collecting the back-reflected light. The intensifier amplifies the light only when the photocathode is ON, thereby producing sectioning of the scene. In the 1990s, the use of active gated image intensifiers coupled to charge-coupled device (CCD) cameras appeared, allowing the development of FLASH light detection and ranging (LiDAR)/laser detection and ranging (LADAR) technology [14], [15]. When LiDAR is used with a narrowband light source, optical modulators, such as Kerr or Pockels cells [16], may be used. To meet the switching speed requirements, these electro-optical modulators must be powered by high-voltage sources, in the order of tens of kilovolts for the Kerr effect cell or kilovolts for the Pockels cell. In addition, the light must pass through a polarizing filter [17], resulting in a loss of active signal.

2) Solid-State Integrated Shutters: In 1996/1997, a CMOS iToF detector, often referred to as a photonic mixer device [18], was proposed. Responsive to a pixel clock, the photocharges generated in the silicon are steered to one of multiple—typically two—collection taps depending on their arrival time. Similar CCD variants had been proposed even earlier in [19] and later in [20].

An advantage of steering converted photoelectrons within a pixel is that it can be more optically efficient than blocking light with an external shutter. For example, if $V_{\max}$ is the maximum signal that can be accumulated in a pixel when the shutter is always open, then $[0, V_{\max}]$ is the range of values that can be generated by a shutter function. However, by steering charges to one of two taps and measuring the difference between the taps, a larger signal in the range of $[-V_{\max}, V_{\max}]$ can be generated, which in most cases of interest improves SNR and performance as well as improving readout chain dynamic range when differential readout is used. Over the years, many different solid-state charge steering shutter mechanisms have been developed [3], [12], [21]–[27] as will be expanded upon in the upcoming sections.

III. AMPLITUDE-MODULATED CONTINUOUS WAVE

Many different kinds of pulse train and shuttering schemes can be used to extract distance. Some of these schemes are highly sensitive to small changes in the laser pulse shape and other circuit variations. Furthermore, for the same optical SNR, they may produce varying accuracy and precision across different distances making their performance hard to predict. By tightly restricting the optical pulse and shutter waveforms, improved predictability can be achieved. One class of pulses and shutters that achieves good stability and predictability is the AMCW method that has now become the most used iToF method.

The optical signal and steering functions in AMCW are sometimes idealized as sinusoidal signals, but in practice,

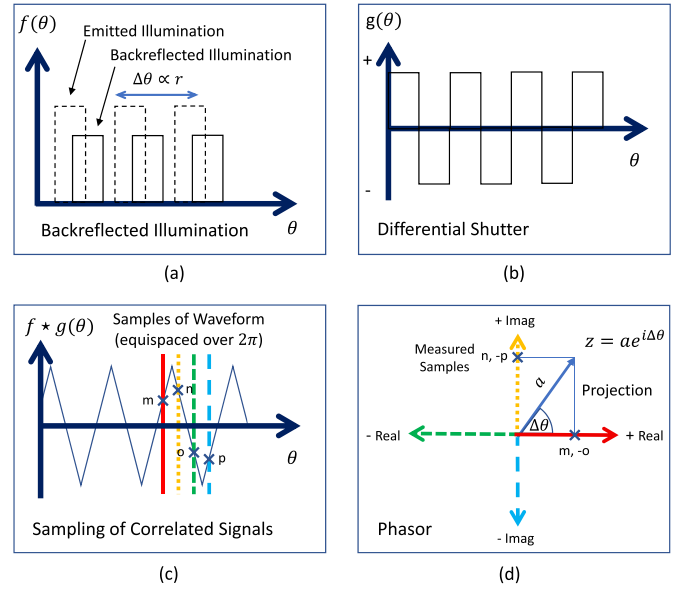


Fig. 2. AMCW measurement formation and sampling. (a) Relationship between emitted and back-reflected illumination. (b) Shutter response. (c) Correlation waveform, where m, n, o , and p are samples at $0^\circ, 90^\circ, 180^\circ$, and 270° . (d) How these samples correspond to projections of the complex phasor encoding amplitude and phase.

the waveforms vary and may be closer to rectangular—they are, however, periodic and have well-controlled fundamental frequencies. While most systems are homodyne, in heterodyne systems such as [28], the optical signal and steering functions may be at different frequencies. In all systems, the optical signal is offset and positively constrained [see Fig. 2(a)] and the shutter may be either positively constrained or differential [Fig. 2(b)], as described in Section II-C. A sensor of the AMCW type is able to measure the correlation of the backscattered radiation with the steering function [Fig. 2(c)], which encodes the range to the object [3].

A. Phase Measurement

Determination of the range of an object using the phase shift of the correlated back-reflected signal very closely parallels the approach taken in phase-shifting interferometry [29]. In general, the most natural representation for an iToF measurement at a particular modulation frequency is as a complex phasor representing the back-reflected illumination waveform at the sensor, $z = ae^{i\Delta\theta}$, where a is the amplitude/active brightness and $\Delta\theta$ is the phase delay. Fig. 2(d) shows that each sample of the correlated signals (matched lines) is equivalent to a projection of the phasor onto a vector whose phase angle is the phase delay of the differential shutter with respect to the emitted illumination. This phasor can be solved as a linear inverse problem from two or more samples of the correlated waveform. In general, the discrete Fourier transform (DFT) is used, acquiring the signal with multiple equispaced shutter phases and integrating over many cycles. Most common implementations include four equispaced captures [3]. Using the four-sample DFT, we find that $z = C(0^\circ) - C(180^\circ) + i(C(90^\circ) - C(270^\circ))$, where $C(\theta)$ is a measurement of the correlated waveform with phase shift of θ .

Once the phase delay is known, the phase can be related to a radial distance by

$$r = \frac{c}{4\pi f}(\Delta\theta + 2\pi n) \quad (2)$$

where the positive integer, n , is the cyclic ambiguity introduced by the sinusoidal nature of the correlated waveform, f is the modulation frequency of the sensor, and c is the speed of light.

Periodic waveforms deviating from pure sine waves give rise to harmonics. While a differential shutter cancels even harmonics in the correlated signals, implementation using digital logic and square waves instead of sinusoids for illumination and gating signals produces higher order odd harmonics that alias to the first harmonic. This leads to a nonlinear range response, sometimes termed “wiggling error” [30]. In the case of four samples, mitigations were proposed using a lookup table (LUT) [31], by phase shifting the reference signal [32], [33], and by phase encoding [33], [34].

B. Multifrequency Systems

The existence of an unknown value, n , means that it is impossible to show the difference between a measurement at 0 m and one at $c/(2f)$ without further information or assumptions, making imaging objects at large distances challenging. As temporal radial distance standard deviation is inversely proportional to modulation frequency, $\sigma_r \propto 1/f$ follows from (2), simply lowering the modulation frequency such that n is always 0 is nonoptimal. By making multiple measurements at different modulation frequencies, it is possible to solve for unwrapped phase as a multifrequency phase unwrapping problem. This scheme retains precision via high average modulation frequency and increases the ambiguous range substantially [28]. However, as the modulation frequency increases, it becomes harder to disambiguate between solutions: a side effect of these wrapped measurements is that mixed pixels and filtration can result in the formation of “flying pixels” [35] at object edges, which can produce both intermediate, also called “webbing” and observed in single-frequency systems, and nonintermediate values or unwrapping errors with values very far from foreground or background.

Approaches such as [36] have been developed to solve the unwrapping problem efficiently. In essence, the methods try to find the combination of integers (wrap orders) that produce a global minimum of a merit function, such as the Euclidean distance between the experimental point and the noise-less representation of phase versus distance. A key aspect is frequency selection, as arbitrarily chosen frequencies tend to have poor properties, and for a fixed ambiguity distance, there is typically a tradeoff between precision/multipath resilience (better at high frequencies [37]) and unwrapping robustness (better at lower frequencies). Approaches such as [38] enable improved modulation frequency choices by allowing explicit optimization for unwrapping robustness. The unwrapping breakdown point typically defines the lowest SNR at which it is possible to range for a multifrequency system, as below this value, the distribution of range estimates becomes multimodal (due to mul-

tiples, noise-dependent solutions for n). Depth processing algorithms are covered in greater detail in Sections V-B and VI.

C. Key Performance Metrics

The precision (σ_r) is an important performance parameter and is readily computed from the SNR of the pixel signal and the modulation frequency (f) [39], with the relationship

$$\sigma_r \propto \frac{\text{SNR}}{f} \quad (3)$$

where, for simplicity, a single modulation frequency is assumed.

Measurement noise can be categorized as: 1) signal shot noise; 2) ambient shot noise; and 3) read noise: these are discussed in Section V-A. A camera system and scene-independent figure of merit (FoM) can be used to characterize the pixel performance depending on the noise regime. When the shot noise from 1) or 2) dominates, the FoM is defined as

$$\text{FoM}_{\text{Shot}} = \frac{\text{MC}_{\text{sensor}} \times \sqrt{\text{QE} \times \text{Pixel Area}}}{f} \quad (4)$$

or when the read noise from 3) dominates, we have

$$\text{FoM}_{\text{Read}} = \frac{\text{MC}_{\text{sensor}} \times \text{QE} \times \text{Pixel Area}}{f} \quad (5)$$

where the modulation frequency-dependent term modulation contrast ($\text{MC}_{\text{sensor}}$) is the probability that a photocharge will generate a signal useful for acquiring depth and higher $\text{MC}_{\text{sensor}}$ produces better SNR. The quantum efficiency (QE) is the probability of collecting a photocharge from a photon.

The frequency selection is usually dependent on the application requirements, e.g., maximum range and phase unwrapping constraints, peak power, and visibility of the typically near-infrared (NIR) modulated light source. Section IV shows the progression of $\text{MC}_{\text{sensor}}$ and QE and how the frequency of operation has been improving consistently over time.

From a system performance perspective, another parameter that is of interest is the MC of the light source (MC_{light}), which refers to how efficient the illuminator is in switching the optical signal ON and OFF. Unless an ideal light source is used, the measured $\text{MC}_{\text{sensor}}$ number reported is typically the product $\text{MC}_{\text{sensor}} \times \text{MC}_{\text{light}}$.

D. Pixel Size Reduction With Increasing Frequency

As with RGB pixels, there is a strong motivation to reduce the size of iToF pixels to both increase the resolution of iToF sensors and to shrink the optical format. Unlike RGB image sensors that are required to produce an esthetic image, ToF pixels are used in metrology applications to produce an objective and precise measurement. As pixels shrink, charge collection is reduced, and hence, the SNR decreases. However, overall performance is also dependent on modulation frequency as shown in (4) and (5). By increasing the modulation frequency, the performance lost to size shrink can be theoretically recovered; this does not come without its challenges as despite the pixel being smaller, MC must be maintained at these higher frequencies both by the pixel and the light source. Other

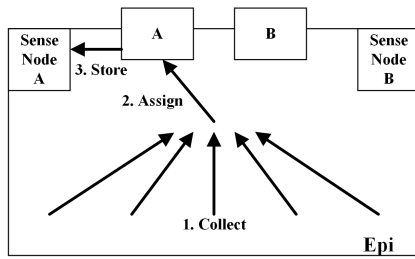


Fig. 3. Demodulation process.

TABLE II
DRIFT FIELD SOURCES OF EACH DEMODULATION PROCESS

Detector types	Collect step	Assign step	Store step
QEM	Gate-induced vertical E-field		Fringe E-field of sense nodes
LEF		Gate-induced lateral E-field	
CAPD		Current-induced lateral E-field	

mitigation techniques include on-demand binning described in Section IV-D.

IV. PIXEL AND CHIP ARCHITECTURES

A. Detector Types

Commonly used iToF detectors can be categorized into QE modulation (QEM) detectors based on photogates, lateral *E*-field (LEF) detectors based on gates, and current-assisted photonic demodulator (CAPD) devices. All these detectors utilize *E*-fields to collect/assign/store photo-carriers based on strong drift-based transport for high-speed demodulation. Fig. 3 shows the three steps of demodulation and Table II shows the sources of *E*-fields of each detector at each demodulation step. In the “collect” step, which is often the most time-consuming due to the thick epi for NIR QE in silicon epi cases, generated photo-carriers drift toward a collection point near the epi front side. A vertically graded doping profile or pinned photodiode (PPD) can be utilized to increase the vertical *E*-field during this phase. As pixel size and power supply voltage shrink, having a vertical doping gradient or PPD becomes crucial—these additional *E*-fields are not included in Table II for simplicity. In the “assign” step, photo-carriers are logically assigned to one particular tap permanently and can henceforth never be reassigned to another tap. This is the step most crucial from a timing perspective to triage the arrival time of photo-carriers and it is imperative that it be executed as fast as possible. In the final “store” step, the photo-carriers are physically stored in a sense node corresponding to the assigned tap.

As mentioned previously, completion of the assign step is a strong indicator of performance, and hence, how fast both “collect” and “assign” are completed is critical to achieving good MC at high modulation frequencies. Because gate-induced vertical *E*-fields are utilized in both the “collect” and “assign” steps in the case of QEM, QEM has the highest MC at the highest frequency among silicon detectors, as shown in

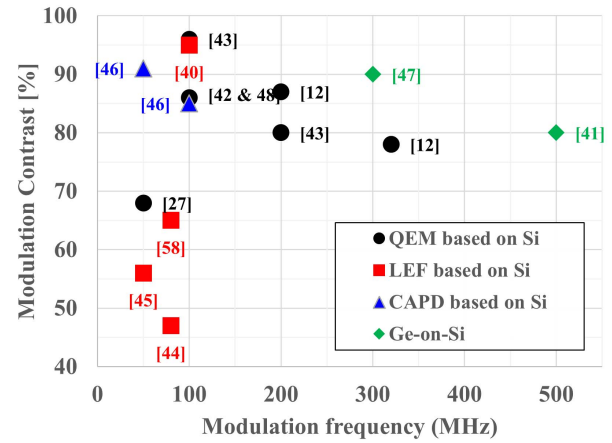


Fig. 4. MC versus modulation frequency.

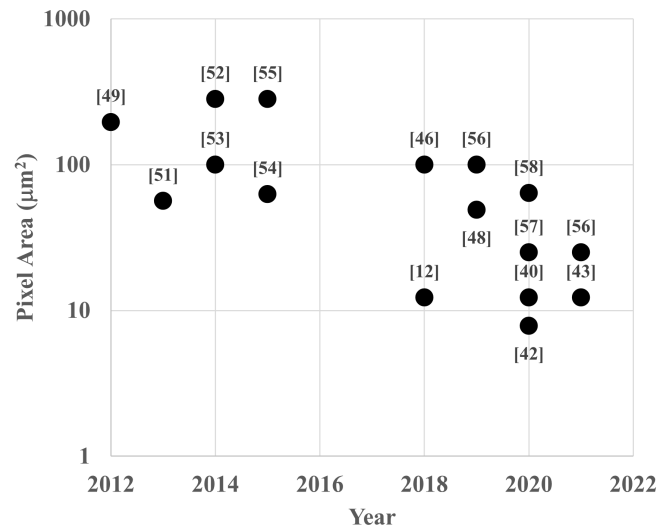


Fig. 5. Pixel size progression since 2012.

Fig. 4 [12]. LEF is expected to have a smaller dark current than other detectors because gates do not have to cover the entire detector area. CAPD makes the detector process simpler due to the absence of gates in the detector; however, a disadvantage of leakage current increasing with shrinking pixel pitch has been reported [40]. Ge-on-Si shows the best MC due to its thin Ge film layer having higher electron mobility than Si [41].

B. Pixel Pitch/Resolution Trend

Figs. 5 and 6 show the pixel size and resolution evolution, respectively, since 2012 for iToF sensors. As can be seen, the race toward a smaller pixel size and higher array resolution has been ongoing for some time. For clarity, only resolutions higher than quarter video graphics array (QVGA) are considered.

In 2018, the notable achievement of the work in [12] was to enable the first megapixel (MP) CMOS ToF sensor at a pixel size of 3.5 μm while maintaining high MC performance at high modulation frequencies. This is now available commercially from Analog Devices [59]. In the past year, two more MP sensors have been published at the

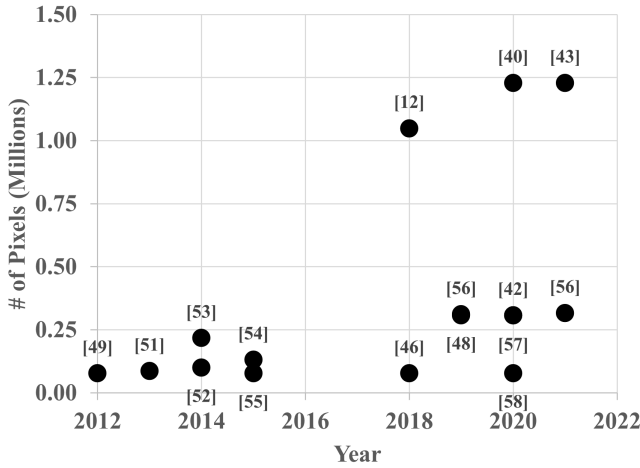


Fig. 6. Array size progression since 2012.

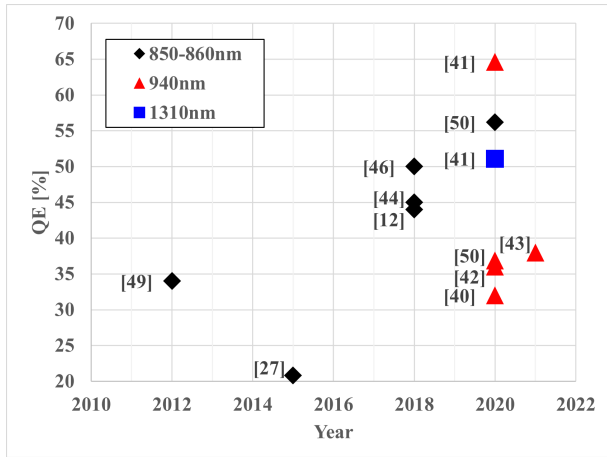


Fig. 7. NIR QE trend of iToF pixels.

same pixel size [40], [43], although they are not yet available commercially.

The tradeoffs with reducing pixel sizes further are discussed in Section III-D. Depending on the application needs, e.g., mobile could benefit from a smaller footprint, the push to get to a smaller pixel size will persist up to a certain level at the cost of performance degradation unless higher frequencies, better filtering techniques, and higher peak lasers are used to mitigate. Even within a particular application, depending on use case, for example, facial recognition could benefit from higher resolution in unbinned mode, while longer range use cases could take advantage of binning to tradeoff resolution with performance.

The highest resolution reported so far is 1280×960 [40], [43]. If pixel sizes are shrunk, there could be a 4K ToF sensor useful for industrial use cases. Appropriate reduction in power consumption is critical for higher resolution arrays.

C. QE Versus Wavelength

Fig. 7 shows the NIR QE trend of iToF pixels. In the case of front-side illumination (FSI) iToF, there is a tradeoff between NIR QE (epi thickness) and MC [27], [44]. While thicker epi

enhances NIR QE, it typically reduces MC because photo-carriers need to be collected from regions farther from the gates. The transition from FSI to back-side illumination (BSI) has enabled NIR QE and MC increases as the pixel shrinks by offering doubled optical path length due to metal reflection and eliminating slow photo-carriers generated from the deep substrate of FSI, respectively [49]. Since BSI was introduced to iToF sensors in [49], patterning at the backside silicon surface of pixels along with deep trench isolation (DTI) has been demonstrated on BSI iToF to yield QE at an 850–860-nm wavelength of above 55% and QE at a 940-nm wavelength of above 30% [40], [42], [43], [46], [50]. The back-surface patterning increases QE by extending the optical path length, while DTI mitigates optical crosstalk caused by this patterning. Ge-on-Si has enabled short-wave infrared (SWIR) QE at 1310 nm as well as QE at 940 nm higher than 60% [41].

D. Multitap and Differential Tap

Different tap architectures have been implemented in iToF systems, including differential tap (two-tap) and multitap (three-tap [60], [61] and four-tap [42], [43], [48], [50]). Multitap architectures have the benefit of reduced motion blur and latency as depth can be computed within a single reset/integrate/readout cycle (if no additional captures are used to correct offsets) for a given modulation frequency. Implementing with only two taps requires more temporal multiplexing, resulting in increased motion blur. A frame buffer is required to store the data until all captures are available for a given pixel and therefore processing cannot start until all captures are read out completely resulting in added latency.

In multitap architectures, pixel offset errors can become a problem. Offsets in pixel measurements can originate from multiple sources, including the amplifier, ADC, pixel clock duty-cycle mismatches, source-follower (SF) mismatches, and the angle of incidence of light falling on the pixel (both ambient and active illumination). These pixel offset errors can be canceled using additional captures, however, this negates some of the advantages of multitap (>2) for low motion blur and low latency, since now, there can be motion blur between the captures and processing cannot start until the captures are fully available. Similarly, multifrequency operation (described in Section III-B) requires multiple captures and can also negate some of the benefits of multitap (>2) operation. Finally, multitap architectures by virtue of having all taps available for readout at the end of integration require a bigger total full-well capacity (FWC; across all taps) making incorporation into small pixels challenging.

E. Pixel Binning

Binning methods provide higher SNR/improved depth performance at the expense of spatial resolution. Making a sensor with small pixels and invoking binning via software configuration at runtime allows a single sensor with small pixels to be used in many different applications requiring different pixel sizes.

Three methods of binning, charge, analog, and digital domain, are compared in Table III in terms of SNR without

TABLE III

SNRS OF SINGLE PIXEL AND 2×2 PIXEL BINNING OF THE PIXEL

	Low photon signal case $N_{ph}^2 \sim N_{SF}^2 + N_{ADC}^2$	High photon signal case $N_{ph}^2 > N_{SF}^2 + N_{ADC}^2$
SNR_{single}	$\frac{S_{ph}}{\sqrt{N_{ph}^2 + N_{SF}^2 + N_{ADC}^2}}$	$\sqrt{S_{ph}}$
SNR_{charge}	$\frac{S_{ph}}{\sqrt{N_{ph}^2/4 + (N_{SF}^2 + N_{ADC}^2)/16}}$	$2 \times \sqrt{S_{ph}}$
SNR_{analog}	$\frac{S_{ph}}{\sqrt{(N_{ph}^2 + N_{SF}^2)/4 + N_{ADC}^2}}$	$2 \times \sqrt{S_{ph}}$
$SNR_{digital}$	$\frac{S_{ph}}{\sqrt{(N_{ph}^2 + N_{SF}^2 + N_{ADC}^2)/4}}$	$2 \times \sqrt{S_{ph}}$

ambient light for simplicity where S_{ph} is the number of photo-carriers integrated during integration time, N_{ph} is the photo-carrier shot noise, N_{SF} is the SF noise, and N_{ADC} is the ADC noise. In charge domain binning, integrated electrons of multiple pixels are added into a shared floating diffusion (FD) and then readout through a single readout path [43]. In analog domain binning, integrated photoelectrons are stored in each pixel's FD and read out through column pixel output lines where the voltages are averaged at the column amplifier [12]. In digital domain binning, each pixel's signal is readout separately and averaged in the digital domain. In the high photon signal case where noise is dominated by photo-carrier shot noise, three of the binning modes yield the same SNR. Assuming FD-limited FWC which is common in iToF pixels, charge binning saturates four times earlier than the rest, while it offers the best SNR in the low photon signal case. Digital binning is the second best for SNR in a low photon signal case; however, it requires four times as many ADC conversions as the other approaches. Analog binning shows the least gain of SNR in low photon signal case, while there is no disadvantage from saturation and the number of ADC conversions.

F. Stacking

A stacking process has been demonstrated in [43] for a relatively small form factor, in which the active pixel array occupies 45.9% of chip area. This suggests that the chip area is not determined by pixel array on the top wafer, but by analog and digital designs on the bottom wafer. Pixel-level Cu-to-Cu bonding two-wafer stacking has been demonstrated with 3.5- μm LEF and 10- μm Ge-on-Si iToF pixels in [40] and [41], respectively. In the case of [40], the pixel-level bonding two-wafer stacking makes the top wafer process simple and MC performance better by minimizing P-well volume in the pixel, which hinders fast photo-carrier collection as pixel size shrinks. In the case of [41], the pixel-level bonding two-wafer stacking enables Ge-on-Si iToF by separating SiGe detector at the top wafer from Si-based in-pixel transistors at the bottom wafer.

G. Device Simulation

Through realistic simulation and modeling, faster design cycles can be achieved at reduced costs. For this, physics-

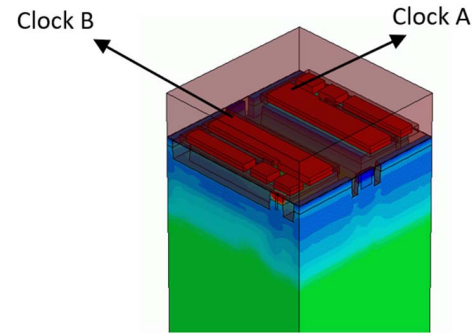


Fig. 8. Simulated 3-D pixel. Only the top portion of the pixel is shown.

based computer-aided design (TCAD) tools are widely used [42], [62]. Since the physics involved is 3-D in nature—particularly so as the pixel size shrinks—3-D modeling is needed [63], [64]. A typical simulation flow involves providing mask layers of pixel lithography steps from a layout editor in the .gds format to a process simulator, which then simulates the various process steps such as lithography, implantation, thermal diffusion, and etching in the same order as would occur in a CMOS flow in a fab. The resulting output structure of the process simulator (Fig. 8) is then used by the device simulator to simulate the electrical behavior. An optical simulation can be performed independent of the process simulation to obtain the distribution of photo-generated electrons in the pixel during the time when the laser light is striking the pixel. During a transient simulation, this distribution is then used in synchrony with the pixel clock signals to model the active illumination of the pixel and the subsequent movement of electrons. The collection of photoelectrons by the gate is then computed to obtain important pixel metrics such as MC, common mode (CM), and QE.

A single iToF pixel is simulated to obtain the pixel metrics QE and MC. Given the long simulation times involved, it is most practical to simulate only a single clock cycle for a two-tap pixel; however, this is generally sufficient to reliably estimate pixel metrics.

H. Analog Architecture

Two key analog functions required for iToF sensors are sensor readout and modulation clock generation for the sensor and illumination. Some iToF sensors capture differential signals and can be read with differential circuits [12], [27]. Differential column amplifiers with good common-mode rejection allow improved on-die interference and power supply rejection and facilitate multigain readout. One downside of differential readout is that single-ended saturation detection may be required since differential readout circuits may interpret deep CM saturation as zero signal.

Methods for measuring and unwrapping phase require multiple samples at different illumination phases and frequencies, as shown in Sections III-A and III-B. In the case of sequential temporal multiplexing of measurements, this places readout requirements comparable to high-frame-rate cameras [65]. The desire for precision depth measurement raises SNR requirements placing pressure on column amplifier and

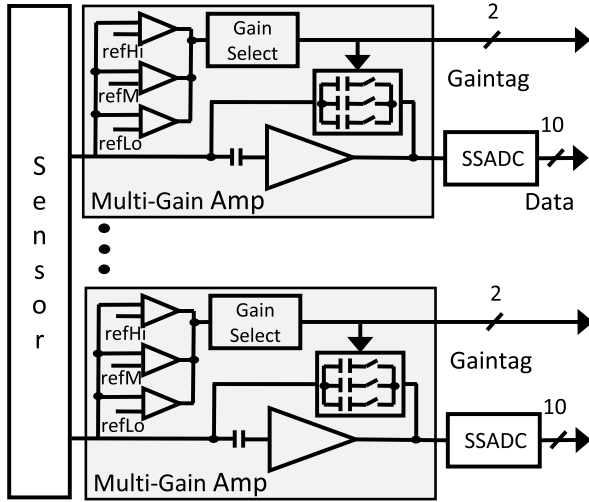


Fig. 9. Multigain column readout.

ADC performance. Dynamic range requirements are further increased by variations in target distance, target reflectivity, and ambient light.

1) Sensor Readout: The requirement of frame rate drives most iToF sensors to use column-parallel readout, and some solutions include readout circuits on both the top and bottom edges of the sensor array [27], [65], [66]. Single-slope ADCs are commonly used with column-parallel readout sensors due to the simple per-column structure and inherently low differential nonlinearity (DNL) [12], [42], [46], [67]. A centrally generated ramp determines the integral nonlinearity (INL) for all the ADCs, while each column ADC measures the crossing of the ramp with the column signal. The main disadvantage of this architecture is that the conversion time grows exponentially with the number of ADC bits.

Multigain column amplifier readout, shown in Fig. 9, can increase the dynamic range while maintaining the number of ADC bits [12]. Two-step or folding ADCs [67] can also increase the number of bits from a single-slope ADC. Finally, column delta-sigma converters show promise for applications requiring higher ADC precision [66], [68], and pixel-level ADCs might drive frame rates higher [69].

2) Modulation Clocks: Errors in clock arrival time at pixels produce errors in computed depth. Calibration has been used to enforce uniform clock arrival time [70]. Providing a modulation clock simultaneously to all the pixels in an iToF sensor presents large instantaneous current demand. Peak current demand can be decreased by spreading the sensor array clocks evenly in time with a delay-locked loop (DLL) [12] or in groups with asynchronous delays [43], [48]. These techniques have resulted in significant reductions in clock power network requirements.

V. NOISE AND DEPTH PIPELINE

A. Random Noise Sources

The most common noise sources for CMOS iToF sensors are system and shot noise [71], which can be modeled as Gaussian and Poisson processes [72], [73]. System noise includes read noise components, such as thermal, flicker,

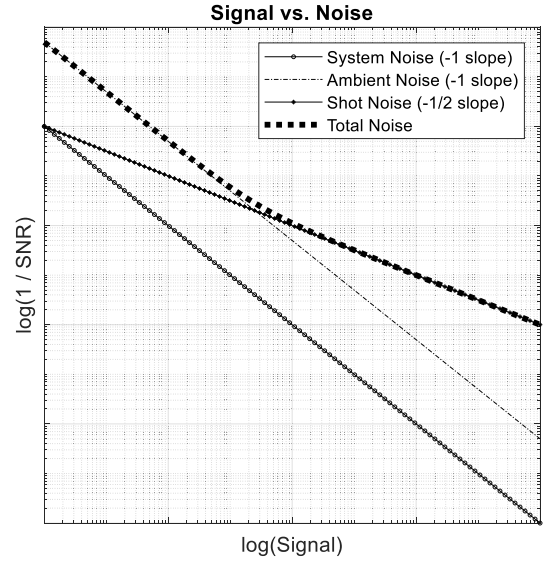


Fig. 10. SNR versus signal for each class of noise, with a total noise plot demonstrating a change in noise regime.

ADC, and charge uncertainty when sampling a capacitor (KTC) noise—typically modeled as Gaussian processes. Shot noise comes from both the active light source and other background/ambient light sources, including dark current; shot noise is Poisson distributed [3] and, at higher signal levels, can be approximated as Gaussian. Using a Gaussian approximation, the SNR can be calculated by the ratio of the signal to the total combined noise, i.e.,

$$\text{SNR} = \frac{S}{\sigma} = \frac{S}{\sqrt{\sigma_{\text{system}}^2 + \sigma_{\text{ambient}}^2 + \sigma_{\text{shot}}^2}}. \quad (6)$$

Fig. 10 shows a log-log graph of these noise sources in isolation as a function of signal level, with arbitrarily chosen ambient light and system noise levels. When the noises are combined, it can be seen that there are different noise regimes where one noise source dominates over another. Both ambient shot and system noise have a slope of -1 ; hence as signal increases, the SNR becomes limited by active shot noise, which has a slope of $-1/2$ due to the underlying Poisson process.

The SNR of a depth measurement, and thus phase noise, is inversely proportional to the signal level, producing a bigger error as the signal decreases [74], leading to worse range precision that ultimately manifests as noisy data, unwrapping errors, and flying pixels. Table IV shows the change in random error—expressed as range temporal standard deviation for a static scene—in each of the three different dominant noises for a $2\times$ change in a system parameter. Depending on the noise regime, the impact of changes such as imaging lens f -number can vary substantially. Ultimately, in order to maximize the SNR and, therefore, decrease the error in the phase calculation, the raw data from the camera are typically spatially filtered, as discussed in Section V-B.

B. Depth Pipeline

Typical depth processing pipelines involve the calculation of complex values from raw measurements, denoising, and

TABLE IV

RELATIVE CHANGE IN RANGE TEMPORAL STANDARD DEVIATION IN RESPONSE TO A 2× CHANGE IN A SYSTEM PARAMETER

Parameter	Range Temporal Standard Deviation		
	System Limited	Ambient Limited	Shot Limited
Distance	4×	4×	2×
ADC Noise	2×	1×	1×
Modulation Frequency	0.5×	0.5×	0.5×
Ambient light	1×	$\sqrt{2}$ ×	1×
Optical Filter Bandwidth	Function of QE, Filter Bandwidth and Ambient Spectrum		
Wavelength			
Reflectivity	0.5×	$\frac{1}{\sqrt{2}}$ ×	$\frac{1}{\sqrt{2}}$ ×
Integration Time			
Q.E.	4×	2×	2×
F/#			
Optical Power / Radiant Intensity	0.5×	0.5×	$\frac{1}{\sqrt{2}}$ ×
MC	0.5×	0.5×	0.5×
Filter Kernel			

phase unwrapping. Additional steps may be required for calibration purposes or to convert the radial distance to Cartesian coordinates.

Several traditional denoising methods for depth data have been proposed in the literature, including the joint bilateral filter [75], adaptive weighted filter [76], median filtering [74], wavelets [77], [78], anisotropic diffusion [79], and total variation [80]. Traditional postprocessing methods used after the depth is calculated have been applied to further denoise the depth data from an image-driven perspective, these include bilateral filters and 3-D block-matching algorithm (BM3D), or data-driven approaches, such as convolutional neural network (CNN)-based algorithms [81], [82]. However, it can be more challenging to denoise after phase unwrapping under low SNR because the unwrapping calculation can itself fail.

The depth processing step most unique to AMCW iToF is that of phase unwrapping, which is critical for applications where the mapping distance exceeds the phase wrapping distance for the raw measurements. Several phase-unwrapping methods exist, which are applied to captures produced either with a single frequency [83], [84] or multiple frequencies [36], [85]–[90]. The methods can be divided into three categories: deterministic methods [36], probabilistic [83]–[85], [87]–[89], or data-driven based on neural networks [90]. There have also been attempts to use neural networks to substitute the whole traditional pipeline by end-to-end pipelines trained with simulated and experimental data [91].

VI. CHALLENGES AND MITIGATION STRATEGIES

A. Mitigating the Effects of Ambient Light

ToF hardware and algorithms are designed to be maximally invariant to any ambient light signal. For example, in a differential pixel, ambient light affects both taps equally and hence to the first order does not affect the differential

signal. Nonetheless, ambient light significantly degrades the performance. It can cause an iToF pixel to prematurely saturate, but more importantly, the noise present in the ambient signal cannot be completely removed. However, the impact of ambient light and its noise can be significantly decreased by the techniques described in this section.

1) Saturation Suppression: Under high ambient light levels, one or more taps of a differential pixel may saturate, thus rendering the pixel data useless; however, in many pixel read-out strategies, only the differential value is used. Background suppression refers to techniques that preserve the differential signal but change the CM so that neither tap saturates. Two background subtraction methods are described.

- 1) In [92], when the value of a tap becomes low, then through matched current sources, an equal amount of current is injected into all taps bringing them closer to the reset value while preserving the differential signal. However, the introduced currents add extra shot noise into both taps, thereby degrading performance.
- 2) In [27], a switched-capacitor circuit resets the CM while theoretically preserving differential signal and accruing KTC noise only in the CM—not in the differential. However, in practice, it is difficult to find space for the required metal capacitors in small pixels and parasitic capacitances degrade the performance, especially for small pixels, causing incomplete CM reset and some loss of differential signal.

In today's small and high-performance pixels, background subtraction is not widely used—possibly because integration times have decreased with the advent of more powerful lasers and shot noise SNR has worsened due to decreased pixel size, thus requiring a greater focus on SNR rather than saturation.

2) Increasing Optical Peak Power: Doubling the light source optical power and halving integration time conserves the total useful signal; however, the amount of ambient light integrated is halved, thereby substantially improving performance. Lasers with high peak power are becoming more readily available [93], [94], enabling this approach. Advantageously, the sensor clock modulation time is also halved, saving chip power. To keep their cost down, these lasers are not designed to deliver high power over long period of time. However, for brief periods of ($\ll 1$ ms), they can deliver large amount of power, provided that they are allowed to cool for periods generally longer than the integration time.

3) Global Shutter: When high peak power lasers are used, integration times shrink, but array readout time often exceeds integration time. A GS is useful to stop the accumulation of ambient light as soon as integration is completed. The pixels can then be read out slowly without being affected by ambient light. Charge domain GS utilizing the FD as storage node for GS operation has been demonstrated with uncorrelated double sampling (DS) [40], [46]. Charge domain GS having a separate storage node per tap for correlated DS (CDS) has been demonstrated previously [12], [42], [43], [48], [95].

4) Restricted Field-of-Illumination: In this mode of operation, the entire field-of-illumination is focused on a reduced integration area [e.g., ten times smaller than the field of view

(FOV)] with correspondingly reduced integration time (e.g., by a factor of ten). Each area is illuminated and acquired time sequentially; the total time to acquire the FOV may be longer because of overheads, but each pixel is integrated for a much shorter period, and thus, the impact of ambient light is reduced. Beam scanning LiDAR is also a variant of this strategy [96], and line scanning has been demonstrated to additionally reduce multipath interference [97].

5) Choice of Wavelength: The solar spectrum measured at sea level is far from blackbody, primarily due to atmospheric absorption—water absorption bands being of particular importance. This means that moderate changes in iToF operational wavelength can have a significant impact on ambient light levels. For silicon-based iToF systems that need to operate outdoors or near building windows wavelengths corresponding to high solar attenuation may be chosen for improved performance in sunlight, even though QE may be less. For example, 940 nm has one-third the ambient light of 860 nm, but QE is typically only 30% less than 860 nm, thus providing a net benefit in high ambient light scenarios.

Some wavelengths beyond the operating capabilities of silicon, such as 1340 nm, have extremely low levels of ambient light. iToF sensors with materials, such as Germanium (Ge) or Indium Gallium Arsenide (InGaAs), have been made to operate at such wavelengths [41], [47], [98]. These benefits must be traded off against the complexity of making devices out of materials that are generally more expensive than silicon and in the case of Ge with large amounts of dark current.

B. Multicamera Interference

Multicamera interference can occur when the illumination from multiple cameras overlaps within the scene, leading to errors in the depth measurements. The simplest approaches to mitigate this problem include separating the cameras in space, time [99]–[101], or by modulation frequency [26], [102]. Code-division multiple access (CDMA) methods have been used to encode the modulation signal with pseudorandom or pseudo-noise (PN) sequences, reducing the constraints of the simple approaches above for applications where the number of cameras and/or their locations are unknown. Direct modulation of the PN sequence has been demonstrated [25]; however, this effectively reduces the average modulation frequency and, therefore, the depth measurement precision [103]. A preferred approach instead modulates the phase of the modulation signal [103], [104], maintaining a high modulation frequency. Different mitigation approaches can also be combined to further enhance the camera performance [105].

C. Multipath Mitigation

Multipath is inherent to any kind of actively illuminated system. Modulated light falling on an object in the FOV may come directly from the camera's light source (the direct path) and also from reflections of the light source from nearby objects (the indirect path). The correct distance measurement corresponds to the direct path and the parasitic reflections from indirect paths typically cause the ToF camera to report a distance larger than the true distance. Neither dToF nor

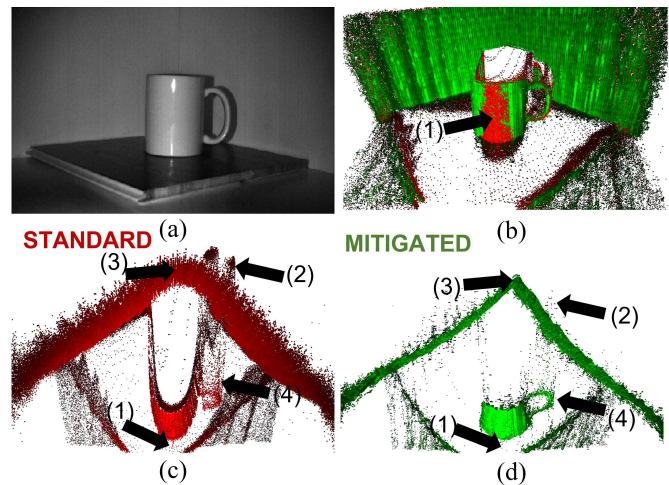


Fig. 11. MP Illustration of the Microsoft proprietary real-time multipath mitigation. (a) Black and White Amplitude Image. (b) Superimposed uncorrected and corrected point clouds. (c) Uncorrected point cloud. (d) Corrected point cloud. 1) No multipath, then red and green point clouds overlap; 2) due to multipath, the standard produces longer distances (red) and corrected mostly mitigates this effect (green); 3) rounded corner (red) versus 90° corner (green); and 4) the handle of the coffee cup is a scattered cloud (red) and solid in the case of the corrected (green).

iToF is free of this effect, especially when the entire scene is simultaneously flood illuminated. dToF perceives the multipath signal as a series of multiple echoes, with the direct path being the first to arrive. The closer in space the direct and the multipath signals the more difficult it is to separate them due to the possible overlapping of the peaks, a phenomenon similar to the Rayleigh criterion for diffraction. In the iToF case, the multipath superimposes sinusoidal waves on each other, resulting in erroneous phase and amplitude estimation [106]. At a system level, preventing multipath interference due to coupling of the illuminator to the imaging sensor is one of the biggest system integration challenges for miniaturized iToF devices and requires careful mechanical design.

Techniques for correcting multipath have been proposed in the literature to correct using structured light [107], [108], ray-tracing methods [109]–[112], closed-form equations or optimization [113], multiple frequencies/spectral methods [114], or by deep learning [115]. Each technique provides different amounts of correction and precision impact and takes from milliseconds to minutes to correct the multipath. Some of the mitigations have been applied in the consumer space by Chronoptics [116], [117] and PMD [118].

Fig. 11 shows Microsoft's real-time multipath correction based on a light pattern illuminator. A glossy white coffee cup is placed close to a corner composed of two white diffuse surfaces (white cardboard) and on top of a polished cherry wood table that it is NIR absorbent and it provides specular for the adequate incidence-collection angles. **Fig. 11(a)** shows a black and white amplitude image, while **Fig. 11(b)–(d)** shows the point cloud from different viewpoints. The point cloud corresponding to the standard processing is shown in red, while the point cloud corresponding to the multipath mitigation enhanced processing is shown in green. Black arrows have been added to show the areas of improvement.

- 1) At the exact front of the cup, there is no multipath, and thus, at that location, both red and green clouds overlap.
- 2) The estimated distance from the standard operation (red) produces (wavy) distances longer than the white cardboards (especially near the corner), while with the enhanced processing (green), the point cloud is flat and close to the cardboards.
- 3) The multipath mitigated cloud (green) recovers the 90° geometry of the corner, without the “bulges” produced by the standard processing (red).
- 4) The handle of the coffee cup where many different types of multipath occur is properly rendered as a solid surface by the enhanced processing (green), whereas the standard processing (red) contains many flying pixels.

D. Motion Blur

Motion blur is an artifact that is produced whenever either the camera, an object in the scene, or both is moving. It can occur because of movement produced either within a single capture, between different captures of the same frequency or between acquiring different frequencies. These motion artifacts produce phase mixing and are most noticeable at the edges of objects, producing errors in phase and brightness. Mitigating motion blur at the hardware level involves the use of GS pixels or short integration time [12]. Algorithms for motion detection and blur compensation have also been proposed and involve modeling the blur and compensating for it [119], [120], introducing a mitigation based on temporal analysis of the depth [120], [121], or detection and compensation of the motion blur at the camera during readout [121], [122].

E. Crosstalk

Crosstalk in CMOS pixels [122]–[125] is caused by two main mechanisms: optical crosstalk and photo-carrier diffusion (electrical crosstalk). These phenomena cause image blur and degrade the SNR; reducing image quality and increasing the number of “flying pixels” in areas of rapid depth changes such as at object edges. As mentioned in Section IV-C, DTI mitigates both optical and electrical crosstalks from a pixel perspective. In addition to the hardware modifications, image processing algorithms can be used to mitigate the blurring effects produced by crosstalk in large-signal gradients.

VII. APPLICATIONS

iToF cameras have huge potential in many consumer and industrial applications, especially those where the real-time 3-D scene, object, or person understanding is required and this cannot be easily achieved through traditional 2-D sensing means. Not only does iToF have great potential, as expressed, but also it has now been productized successfully in many different arenas.

iToF-based object understanding has particular applicability to transportation and warehousing. Combined with artificial intelligence (AI) and robotics, iToF cameras have enabled the digitization, optimization, and automation of freight handling [127]. In manufacturing, retail, and warehousing, improvements in processes and logistics have been proven

through real-time and accurate measurement of the spatial dimensions of products and their volumetric load [128], [129] and inventory tracking [130], [131].

Many robotics scenarios require spatial reasoning about an environment, ToF sensors have been used to enable safe operation and complete situational awareness on autonomous mobile robot (AMR) and automated guided vehicles (AGVs) [132] in warehouses, manufacturing facilities, and even offices. They have been used for specialized tasks, such as forklift pallet pocket detection [133] and on-the-fly programming of painting robots [134].

In augmented reality (AR)/virtual reality (VR) [135], [136], iToF cameras are used to provide the 3-D reconstruction and spatial mapping [137] of the environment to allow the placement of, and occlusion with, virtual objects. The cameras also provide articulated hand tracking/gesture control [138] of the user interface and manipulation of virtual objects—techniques reviewed by Li *et al.* [139]. High-value applications include medical imaging visualization and AR-assisted surgery [140].

The automotive industry has primarily focused on in-cabin applications using iToF, including gesture control of the entertainment system, occupant detection, and driver monitoring [141], [142]. Other people understanding applications include estimating the pose of a person or, “body tracking,” include in gaming [143], through to personalized fitness workouts [144].

Key to the uptake of depth technology is incorporation into high-volume commercial and consumer products. IoT applications, such as people counting for smart buildings [145] and patient fall detection [146], [147], are good examples of the potential in the technology. In mobile phones, front-facing sensors enable secure user identification (including payment systems) and advanced photography, including fast focusing and bokeh postprocessing, and blurring out the background in real-time video, while rear-facing sensors facilitate advanced photography/video and AR applications. Fusion of depth with high-resolution passive images also enables guided upsampling [148], reducing resolution requirements for the depth sensors.

Android mobile phone manufacturers have incorporated iToF sensors into select devices (including Samsung S10 5G, A80, S20+, S20 Ultra, and Note 10+; Huawei P30 Pro and Honor View 20; LG G8 ThinQ and V60; Sony Xperia 1 II and III; Oppo Rx17 Pro; and Vivo Nex, Meizu 17 Pro), with either front or rear orientation, or in some instances using two iToF sensors or a rotating sensor to cover both orientations. iToF usage in these devices have been met with limited adoption to date, however, the inclusion of depth cameras in the Apple iPhone 12 Pro (a dToF rear camera in addition to a structured light front-facing camera [149]) is expected to lead to rapid growth in the mobile phone depth camera market [150].

The most successful applications to date utilize high-resolution depth data, operating over relatively short distances of 1–10 m and are often situated indoors, or in environments with limited ambient light. As the technology continuously evolves—especially with regard to smaller sensor size discussed in Section IV-B and lower power consumption—we expect more uptake of the technology by wearable and new

device form factors where portability and miniaturization are key. Improvements in ambient light resilience discussed in Section VI-A enable a variety of exterior automotive applications, including parking assist (autonomous parking) and close-range navigation, as well as use-anywhere depth-enabled portable devices. The use of ToF cameras in challenging scenes containing higher levels of multipath, such as industrial environments containing polished floors or metal surfaces, is made possible using mitigation techniques described in Section VI-C, providing improved performance and enabling further deployment for new applications operating in these spaces.

Ultimately, we expect a future of ubiquitous depth, where depth sensing is no longer seen as an exotic technology, but something deeply embedded in the life of the average person, through devices such as cars, phones, smartglasses, and unspoofable depth-enabled doorbells.

VIII. CONCLUSION

iToF design brings together a rich set of disciplines spanning device physics, device simulation, unique CMOS pixel design, specialized analog readout chain design, optics, and light source design through signal processing algorithms. The technology has progressed significantly over the years and today provides the compactness and performance necessary for many applications. iToF technology now provides small pixels, high XY resolution, high Z precision, and accuracy—all using a foundry process similar to the familiar CIS process.

This article focuses on the robust AMCW method using iToF CMOS pixels. In AMCW iToF, the key pixel performance drivers are MC, modulation frequency (f), and QE. Device simulation of pixels, an important aspect of pixel design, is described. Multifrequency methods extend the useful unambiguous operating range while retaining system performance by maintaining a high average modulation frequency. We also show the effect on the performance of system-level parameters (e.g., lens $F\#$ and integration time) and scene parameters (e.g., object reflectivity). Different iToF pixel types are listed and the charts for major trends over time for pixel size, MC, frequency, NIR QE, and sensor array size are provided. Various forms of pixel binning that trade off XY resolution for total system power are explained. Readout chain analog requirements and techniques are listed. State-of-the-art filtering and processing algorithms that substantially enhance the quality of the acquired data are described. iToF is not without weaknesses, but many potent mitigation strategies to overcome these are described. iToF technology has already found significant use in the AR-VR/industrial/consumer-fitness/commercial fields, and as the technology progresses, the space of scenarios that can be served is rapidly expanding.

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